

Secret Codes & Files

Pre-Assessment 2 · 9 questions · 30 points

Name: _____ Date: _____ Score: / 30

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write **“not yet”** and move on.

Work on your own, without a computer. You have about **40 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Letters are numbers

3 points

Use these facts: `ord("A")` is 65 and `ord("a")` is 97. Letters are numbered in alphabet order.

Expression	Result
<code>ord("C")</code>	
<code>chr(70)</code>	
<code>chr(ord("b") + 2)</code>	

2. Around the alphabet

3 points

(a) What is $27 \% 26$? (1 pt)

(b) What is $(23 + 5) \% 26$? (1 pt)

(c) Number the letters $a = 0, b = 1, \dots, z = 25$. Which letter do you get if you shift `y` forward by 3 and wrap around? (1 pt)

3. Secret Agent Cipher Lab

4 points

```
def shift_letter(ch, shift):
    pos = ord(ch) - ord("a")
    new_pos = (pos + shift) % 26
    return chr(new_pos + ord("a"))

def encrypt(text, shift):
    result = ""
    for ch in text:
        if ch == " ":
            result = result + ch
        else:
            result = result + shift_letter(ch, shift)
    return result
```

(a) What does `encrypt("hi zoo", 2)` return? (2 pts)

(b) What does `encrypt("jk bq", -2)` return? (1 pt)

(c) A message was encrypted with a shift of 5. What shift would you pass to `encrypt` to decrypt it? (1 pt)

4. Crack the code

3 points

Use the `encrypt` function from Question 3. You intercepted a secret word but don't know the shift.

```
secret = "khor"
for shift in range(_____):
    print(shift, encrypt(secret, _____))
```

(a) Fill in the blanks so the loop tries every shift from 1 to 25 and decrypts with it. (2 pts)

(b) Which shift reveals a real English word, and what is the word? (1 pt)

5. Story Reader

4 points

FILE: PETS.TXT

```
cat
dog
parrot
```

```
with open("pets.txt") as f:
    lines = f.readlines()
print(len(lines))
print(lines[1].strip().upper())

count = 0
with open("pets.txt") as f:
    for line in f:
        if "a" in line:
            count = count + 1
print(count)
```

(a) Write exactly what the program prints.

(3 pts)

(b) What does `f.read()` give you? Circle one.

(1 pt)

- ☐ A A list of lines ☐ B One string holding the whole file
- ☐ C Only the first line ☐ D The number of lines

6. Where is the file?

2 points

FOLDER

```
KidCode/
  story_reader.py
  books/
    fox.txt
```

(a) `story_reader.py` runs from the `KidCode` folder. Which line opens `fox.txt`? Circle one.

(1 pt)

- ☐ A `open("fox.txt")` ☐ B `open("books/fox.txt")`
- ☐ C `open("KidCode.fox.txt")` ☐ D `open("books.fox")`

(b) What error would `open("fox.txt")` cause?

(1 pt)

7. Diary Logger

4 points

```
with open("log.txt", "w") as f:
    f.write("Day 1\n")
with open("log.txt", "a") as f:
    f.write("Day 2\n")
with open("log.txt", "a") as f:
    f.write("Day 3\n")
```

(a) What is inside `log.txt` after the program runs? (2 pts)

(b) If the **third** block used `"w"` instead of `"a"`, what would `log.txt` contain? (1 pt)

(c) If none of the `write` calls had `\n`, what would `log.txt` contain? (1 pt)

8. Add up a file

4 points

`numbers.txt` has one whole number on each line. Write a function `total_from_file(filename)` that opens the file, adds up all the numbers, and **returns** the total.

FILE: NUMBERS.TXT

```
4
10
7
```

For this file, `total_from_file("numbers.txt")` returns 21.

9. Diary entry

3 points

Write a function `add_entry(text)` that adds `text` as a **new line** at the end of `diary.txt`, without erasing what is already there.

END OF ASSESSMENT